

Indian Institute of Information Technology Sri City, Chittoor

CCN – Spring 2024

TE-1 Answer Keys

Duration: 90 Minutes

Maximum Marks: 20

1	a) Cookies/HTTP Cookies/web cookies b) i) 8 RTT0 ii) 14 RTT0	$1M + 2M + 2M = 5M$
2	a) i) $2*2+4*2+2*8 = 28$ bits ii) $150k/2 = 300k/4 = 600k/8 = 75000$ frames/s iii) $1/75000 = 13.33$ micro seconds iv) $2*150+2*300+2*600 = 2100$ kbps OR iv) $(2*2+4*2+2*8)*75000 =$ kbps NOTE: Please carefully check 'b', 'B', and 'k'. b) Application Layer Transport Layer Network/Internet Layer Data Link Layer Physical Layer c) L2: Physical Layer, Data Link Layer L3: Physical, Data Link, Network	$1M*6 = 6M$
3	a) $L/R1 + L/R2$ b) The delay components are processing delays, transmission delays, propagation delays, and queuing delays. All of these delays are fixed, except for the queuing delays, which is variable.	$2M + 2M = 4M$
4	a) HTTP server sends requested files to clients without storing any state information about the client. If a particular client asks for the same object twice in a period of a few seconds, the server does not respond by saying that it just served the object to the client; instead, the server resends the object, as it has completely forgotten what it did earlier. Because an HTTP server maintains no information about the clients, HTTP is said to be a stateless protocol. b) Application Layer: • Consists of user programs, network applications that does work at hand • Examples: • File transfer, Remote login, Mail, Web access • Protocols: FTP, Telnet, Simple Mail Transfer Protocol(SMTP), HTTP.	$1M + 3M + 1M = 5M$

Presentation Layer:

- Concerned with syntax and semantics of information transmitted
- Translation
- Encoding data: Data compression/conversion, encryption and decryption

Session Layer:

- Allows to establish a session between peers
- Dialogue control: Session can allow bidirectional traffic or only unidirectional traffic.
- Token management: In some protocols, it is required that both sides do not attempt same operation at same time. Session layer provides tokens to perform such actions
- Synchronization: Pausing and resuming a download.

Transport Layer:

- Connection-oriented services to applications
 - flow control
 - guaranteed delivery of messages to destination
- Ensures data delivery is
 - error-free
 - in sequence
 - no loss, duplication and corruption of packets

Network Layer:

- Interface between host and network
- Routing
- Congestion and deadlock
- Internetworking

Data Link Layer and Physical Layer:

- **Data-link layer**

- Takes packet from network layer and moves it to the next router
- error-free delivery: computes error detection information

- **Physical layer**

- Controls transmission into the network cable.
- Defines electrical signals.

c) What is the motivation behind studying CCN?

Answer: The computer, communication, and networks (CCN) is core for telecommunications. It tells us basically the concept of the information transformation from sender to receiver via communication channels. It sounds interesting to know about transformation of the information via telephone lines, coaxial cables, optical fibers or antennas right. Further, we know that structure/unstructured data is communicated in forms of packets using several protocols and then collected. It is just a subject that let you know how all the telecommunication processes occur. Furthermore, knowing a computer network can help you in more understanding the ability of your concepts. That's making your telecommunication way more interesting. In addition, resource sharing such as printers, data storage devices, and other hardware and software resources are included that can help to reduce costs, increase efficiency, and improve productivity.